INCH-POUND
MIL-DTL-24643/71B
27 November 2023
SUPERSEDING
MIL-DTL-24643/71A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 1,000 VOLTS, FLEXIBLE, TYPE LSSCF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, tin-coated (see table I for size). Conductors in accordance with ASTM B174, class K, for 10 AWG and smaller, and ASTM B172, class K, for 8 AWG and larger. For rope stranded conductors, the minimum number of rope groups shall be nineteen.
Second	-	A separator may be used at manufacturer's option when required to provide free-stripping insulation.
Third	-	Thermoset insulation, white (see table I for diameter).
Fourth	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/71	Type and size	Cold bending mandrel diameter max. (±0.0625 inches)	Conductor size (AWG/MCM)	Number of strands (nominal)	Conductor resistance (ohms)	Insulation resistance (megohms)	Insulation diameter min. (inch)	Jacket thickness min. avg. (inch)	Cable diameter (inches)	
									min.	max.
-01UN	LSSCF-3	0.75	16	26	4.624	500	0.110	0.012	0.135	0.150
-02UN	LSSCF-4	1.00	14	41	2.88	500	0.155	0.012	0.180	0.195
-03UN	LSSCF-10	1.25	10	105	1.14	500	0.195	0.017	0.230	0.245
-04UN	LSSCF-16	1.50	8	168	0.729	500	0.270	0.022	0.320	0.340
-05UN	LSSCF-26	1.75	6	259	0.459	250	0.315	0.022	0.365	0.385
-06UN	LSSCF-42	2.25	4	420	0.289	200	0.370	0.032	0.440	0.460
-07UN	LSSCF-66	2.50	2	665	0.184	150	0.435	0.042	0.525	0.555
-08UN	LSSCF-105	3.25	0	1,045	0.115	125	0.555	0.047	0.655	0.685
-09UN	LSSCF-133	3.50	00	1,330	0.0922	100	0.605	0.047	0.705	0.735
-10UN	LSSCF-165	4.00	000	1,672	0.0731	100	0.680	0.047	0.780	0.820
-11UN	LSSCF-210	4.25	0000	2,109	0.0580	90	0.730	0.047	0.830	0.870
-12UN	LSSCF-250	4.50	250 MCM	2,449	0.0491	80	0.830	0.047	0.930	0.970
-13UN	LSSCF-300	4.75	300 MCM	2,989	0.0409	75	0.860	0.047	0.960	1.010
-14UN	LSSCF-350	5.00	350 MCM	3,458	0.0354	75	0.890	0.047	0.990	1.040
-15UN	LSSCF-400	5.25	400 MCM	3,990	0.0310	70	0.960	0.047	1.065	1.115
NOTE: Minimum bend radius of the cables is the listed mandrel diameter in table I for the cold bending test.										

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand, conductor to ground (volts, root mean square, min.)	5,000
Insulation resistance, conductor to ground (megohms/1,000 feet, min.)	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Abrasion resistance test	Not applicable

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Cold bending (cable)(tested at -4 °F [-20 °C], mandrel diameter in accordance with table I)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	250
Jacket (cable)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Jacket (cable)	

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Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	75

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Insulation	0.2
Insulation (durometer) – type A (hardness, min./max.)	60/75
Jacket (durometer) – type A (hardness, min./max.)	60/75
Smoke index (max.)	
Jacket	25
Insulation	35
Toxicity index (max.)	
Jacket	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Electrical moisture absorption	No failure
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSSCF (3 through 66)	1,000
LSSCF (105 through 400)	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-087)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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